

The Leviathan in the Boardroom

Evidence on Industrial Policy by State Acquisition

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This research proposal focuses on industrial policy conducted by state equity participation, a ubiquitous yet understudied type of state intervention. It outlines preliminary results on post-war Italy (1945-73), obtained by leveraging a panel of more than thirty thousand joint-stock firms, covering more than 90% of the total equity in the universe. The balance sheets of firms are linked to ownership information from the three main state holdings: at peak, state-invested firms accounted for over 20% of total assets in the database. Complementary Difference-in-Differences models show that state participation increases firms' assets and equity, but has heterogeneous effects on profitability, depending on pre-acquisition performance. Importantly, sector-level estimates show positive spillovers to privately owned firms, consistent with state investment acting as a coordination device for private capital. However, clearer evidence requires addressing current data limitations. I plan to strengthen the dataset by digitising archival sources, including balance sheets for missing years, acquisition timing and extent, unrealised acquisitions, input-output tables to measure spillovers, and historical patent data. This will enable a more rigorous estimation of the causal effect of state acquisition on firms' economic performance, accounting for spatial and sectoral linkages à la Lane (2025). Two further lines of inquiry arise: explaining the causes of acquisition, with particular attention to political connections based on board compositions; and shedding light on the impact of acquisitions on the innovativeness of participated firms, tracking registered patents. Concluding, I suggest further avenues of research into French *dirigisme* based on yet unexplored archival data sources. In a moment when several governments are expanding or planning to expand their equity stakes in selected firms, this research will provide timely and policy-relevant insights on whether and when they are likely to achieve their aims.

1 Introduction

President Trump's recent acquisition of a 10% equity stake in chipmaker Intel was referred to as historic and "pretty darn bonkers".¹ Yet, the US Government has acquired shares of multiple key domestic firms in recent years (US Steel, Lithium Americas, MP Materials). In

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¹Financial Times, [Aug 25, 2025](#).

pursuit of what has been defined “technological sovereignty with American characteristics”,² the US administration has fostered “this provocative idea that the government should get a stake or some kind of say in the companies that it is helping, in one way or another”.³ Similarly, the UK government will empower the National Wealth Fund and the British Business Bank to take significantly larger and “more strategic” equity positions in domestic firms to “crowd in private finance” and “anchor future superstar firms in the UK” (UK Government 2025, pg. 51). This phenomenon might soon extend to Europe, where countries are scrambling to rebuild industrial pipelines for green technologies (Veugelers, Tagliapietra, and Trasi 2024) and defence (Wilkinson 2020), a sector dominated by large state champions partially owned by governments.

These acquisitions are justified mainly by two arguments: the need to promote infant industries (Juhász 2018; Juhász, Squicciarini, and Voigtländer 2024; Krueger and Tuncer 1982), good jobs (Dani Rodrik 2022), and national security concerns. While this topic has drawn renewed academic attention (Rosenstein-Rodan 1943; Nurkse 1953; Lane 2019; Juhász, Lane, and Rodrik 2024), causal evidence on how state equity stakes operate as industrial policy tools remains scarce (Breznitz and Gingrich 2025; Curi, Mancuso, and Scarpa 2025).

I would like my postdoc to be devoted to studying State equity acquisitions for industrial policy purposes, centering my inquiry on the case of Italy’s state holdings (1945-1975). I will now outline the preliminary results I have so far, and later describe my plan to enhance our understanding of state acquisitions more broadly.

2 Preliminary Results

Industrial policy encompasses diverse interventions aimed at the transformation of the economic structure in pursuit of some public interest (Juhász, Lane, and Rodrik 2024). Its success depends on credibility, as the fear of policy reversals discourages private investment (Juhász and Lane 2024). One solution is to delegate policy to insulated bureaucracies (Bersch and Fukuyama 2023; Besley et al. 2022).

This describes 20th-century Italy (Gasperin 2022; Monfardini, Quattrone, and Ruggiero 2022), where the government created three state holdings — IRI, ENI, and EFIM - to modernize the economy and reduce reliance on imports (Amatori 2020). Between the 1950s and 1970s, Italy experienced its “economic miracle” (Toniolo 2013), while state holdings controlled between 17.9% and 29.7% of the capital of all Italian joint-stock firms (Toninelli and Vasta 2011, pg. 83). Despite the scale and apparent success of these holdings, few quantitative studies exist (e.g., Vannini 2017). To the best of my knowledge, none assess the causal impact of coordinated state acquisitions on firm or sector performance.

²Financial Times, [Oct 12, 2025](#).

³New York Times, [Sep 2, 2025](#).

To fill this gap, I merge two data sources on firms’ ownership and performance after World War II. The ImitaDB dataset (Giannetti and Vasta 2006) provides balance sheet data for Italian joint-stock firms, while Toninelli and Vasta (2011) identify 669 state-invested firms through corporate ownership mapping. Unfortunately, both sources suffer substantial limitations, which I describe in Section 3.1 together with my plans to address them during my postdoc.

Preliminary Difference-in-Differences analyses (Arkhangelsky et al. 2021; Callaway and Sant’Anna 2021; de Chaisemartin and D’Haultfœuille 2020) indicate that state participation increases firm assets and equity, although profitability effects vary: well-performing firms maintain their ROA, while struggling firms stagnate. This suggests that state investments ease financial constraints but may dilute profit-maximisation. This supports the view that public acquisitions should be employed to correct coordination failures (Juhász, Lane, and Rodrik 2024), provided such gains outweigh reduced profitability. Sector-level results mirror this pattern, and notably, fully private firms in state-targeted sectors also grow, hinting at positive coordination effects rather than simple derisking. Importantly, these private firms see an improvement in profits as well: this hints that state acquisitions may have its best effects on neighbours, who receive a boost in activity without sacrificing effective management.

While these results are already suggestive, stronger evidence requires addressing the weaknesses of current data sources.

3 Research Plan

3.1 Data Production

I will enhance the empirical base by expanding firm-level data, detailing acquisition timing and extent, identifying unrealized acquisitions, and adding input–output tables to measure inter-sectoral spillovers. Each point highlights the theoretical grounds for the inclusion of the data and its precise source.

Balance sheet data (1945–1983) ImitaDB maps firms in four post-war benchmark years, and then traces their balance sheets back to the previous benchmark. This approach leads to missing firms active between two benchmark years. Plus, it does not backtrack firms mapped in 1983, creating a 10-year gap. This limits the timespan of the study and hinders analysis of how acquisitions affect firm survival. In collaboration with Prof. Michelangelo Vasta (Siena), I plan to digitize missing years using *Italian Joint-stock Companies: Statistical Notes* at the National Library in Rome. If acquisition truly increases the likelihood of firm exit, without this data integration selection on the dependent variable would underestimate this effect, positively biasing results. Additionally, this pipeline will allow the integration of balance sheet data on firms’ debts and debt service, which would allow us to gauge dynamics of investment pre- and post-acquisition.

Acquisition details (1945–2001) Ownership data is available only for benchmark years and for the holdings’ foundation year. This masks the true beginning and end dates of state shareholding, and largely ignores changes in its extent. Digitized archives of state holdings contain attachments to yearly consolidated balance sheets that record these changes. Precise dating is crucial to test pre-trends and exits, i.e. the “ability to let losers go” which is crucial to successful industrial policy (Juhász, Lane, and Rodrik 2024, pg. 218). Moreover, as Bortolotti, Fotak, and Wolfe (2022) show in more recent years, minority and majority stakes have opposite effects on innovation, requiring detailed coding of stake size.

Unrealized acquisitions (1945–2001) Four-year planning documents from the same holding archives record potential but unrealized acquisitions. These cases provide valuable counterfactuals that mitigate selection bias (à la Bena and Li 2014). Clearly, unrealized investment will not have been carried out for reasons that might correlate with outcomes. However, comparing the size of the effect estimated with this restricted control group can give a clearer idea of the impact of selection bias.⁴

Input–output tables (1964–1975) Tables from the *Italian Statistical Bulletins* will allow measurement of spillovers through sectoral linkages. While within-sector competitors seem to benefit, increased firm size could also harm upstream suppliers or downstream buyers by generating dominant positions. It is thus crucial to add a thorough analysis of the ripple effects of the policy on supply chains, in order to disentangle the extent to which the profits observed are due to weaker competition.

3.2 Analyses on Linkages, Innovation, and Political Connections

This data work will open several research avenues. First, it will underpin a strong job market paper on the effect of industrial policy through state acquisition. Remarkably, the study would offer the first causally identified account of this widespread yet understudied mode of state intervention in the economy. Beyond contributing to this rapidly growing literature, the paper would speak to the connection between state-owned enterprises and structural transformation. Excluding the Chinese experience (e.g. Huang et al. 2017), this literature offers little systematic evidence on whether and when SOEs have promoted the structural transformation of advanced economies (Cardinale, Landoni, and Mi 2024).

Furthermore, I plan to research the impact of state acquisitions on the innovativeness of firms in a joint work with Professor Vasta, who has extensively researched innovation using historical economic data. We will leverage already digitised patent data from the Italian State Archives and from the US Patent and Trademark Office to inquire whether state intervention can foster innovation. While recent work has highlighted the durable impact of knowledge transfers

⁴On a general note on the matter of self-selection, it is important to mention that no randomly assigned industrial policy exists in the wild, which makes ATT estimates likely to be more informative than ATE ones.

(e.g. in Italy: Giorcelli 2019; and in China: Giorcelli and Li 2021) and the value of government-funded R&D (Gazzani et al., n.d.) we have little systematic evidence on the impact of state control on innovation (Falck, Gollier, and Woessmann 2011). As anticipated in Section 1, this question is particularly relevant for the development of green transition technologies in Europe (Veugelers, Tagliapietra, and Trasi 2024; Semrau, Peterson, and Meissner 2060).

Political connections are frequently cited as reasons for failed state intervention (Faccio, Ma-sulis, and McConnell 2006), especially in relation to misallocated credit (Khwaja and Mian 2005). Faccio and McConnell (2023) examine post-war Italy and show that their quasi-random disruption leads politically connected firms to underperform. It is thus paramount to explore the impact of political connections on state holding acquisition decisions. The data, which features a full time-varying mapping of the members of the boards of administrators and syndics, would allow me to proxy the distinction between political and economic acquisitions, leveraging directories of politicians to identify which ones served on corporate boards. Moreover, the location of firms yields a proxy of the electoral expediency of acquisitions, whose occupational impact might have been targeted at swing constituencies. Often, policies are implemented privileging political benefits at odds with their stated economic objectives (e.g. Colussi et al. 2020). Such a project would help understanding what the warning signs are.

Finally, I will coordinate with Professor Vasta the release of an updated version of the ImitaDB dataset, with all years from 1945 to 1983 completely mapped. ImitaDB is currently the most comprehensive source of historical firm-level information in Italy. The improvement outlined in Section 3.1 would position it as the go-to source for any quantitative study on the evolution of the Italian economy. Over the longer horizon, I will broaden the scope of my research on State-led industrialisation to France. The National Archives offer detailed biographical and academic data for all graduates of the École Nationale d’Administration from 1955 to 1978. While *dirigisme* is historically credited with having fostered growth, no quantitative study has yet investigated the success of this bureaucratic control strategy nor its potential replicability. The fact that it focused on the selection of aligned decision-makers, rather than on outright state control, might help address the clash between political and economic incentives arising in a system of state holdings.

I believe this research plan has the potential to generate rigorous contributions worthy of leading international journals and shed light on a ubiquitous, consequential, and understudied aspect of state economic intervention.

4 References

- Amatori, Franco. 2020. “IRI: Financial Intermediary or Entrepreneurial State?” *Financial History Review* 27 (3): 436–48.
- Arkhangelsky, Dmitry, Susan Athey, David A. Hirshberg, Guido W. Imbens, and Stefan Wager. 2021. “Synthetic Difference-in-Differences.” *American Economic Review* 111 (12): 4088–118.
- Bena, Jan, and Kai Li. 2014. “Corporate Innovations and Mergers and Acquisitions.” *Journal of Finance* 69 (5): 1923–60.

- Bersch, Katherine, and Francis Fukuyama. 2023. "Defining Bureaucratic Autonomy." *Annual Review of Political Science* 26 (Volume 26, 2023): 213–32.
- Besley, Timothy, Robin Burgess, Adnan Khan, and Guo Xu. 2022. "Bureaucracy and Development." *Annual Review of Economics* 14 (Volume 14, 2022): 397–424.
- Bortolotti, Bernardo, Veljko Fotak, and Brian Wolfe. 2022. "Government Share Ownership and Innovation: Evidence from European Listed Firms." {{SSRN Scholarly Paper}}. Rochester, NY: Social Science Research Network.
- Breznitz, Dan, and Jane Gingrich. 2025. "Industrial Policy Revisited." *Annual Review of Political Science* 28 (Volume 28, 2025): 329–50.
- Callaway, Brantly, and Pedro H. C. Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics*, Themed Issue: Treatment Effect 1, 225 (2): 200–230.
- Cardinale, Roberto, Matteo Landoni, and Zhifu Mi. 2024. "Global State-Owned Enterprises in the 21st Century: Rethinking Their Contribution to Structural Change, Innovation, and Public Policy." *Structural Change and Economic Dynamics* 68 (March): 468–72.
- Colussi, Tommaso, Giampaolo Lecce, Marco Manacorda, and Massimiliano Gaetano Onorato. 2020. "The Politics and Economics of Government Aid: Evidence from the Italian Cassa Per Il Mezzogiorno."
- Curi, Claudia, Paolo Mancuso, and Alessandro Scarpa. 2025. "State-Owned Enterprises: A Bibliometric Review and Research Agenda." *Finance Research Letters* 74 (March): 106749.
- Dani Rodrik. 2022. "An Industrial Policy for Good Jobs." The Hamilton Project, Brookings.
- de Chaisemartin, Clément, and Xavier D'Haultfœuille. 2020. "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review* 110 (9): 2964–96.
- Faccio, Mara, Ronald W. Masulis, and John J. McConnell. 2006. "Political Connections and Corporate Bailouts." *The Journal of Finance* 61 (6): 2597–2635.
- Faccio, Mara, and John J. McConnell. 2023. "Political Connections Cause Resource Misallocation: Evidence from the Fall of Fascism in Italy." {{SSRN Scholarly Paper}}. Rochester, NY: Social Science Research Network.
- Falck, Oliver, Christian Gollier, and Ludger Woessmann, eds. 2011. *Industrial Policy for National Champions*. The MIT Press. <https://www.jstor.org/stable/j.ctt5hhg19>.
- Gasparin, Simone. 2022. "Lessons from the Past for 21st Century Systems of State-Owned Enterprises: The Case of Italy's IRI in the 1930s." *Structural Change and Economic Dynamics* 62 (September): 599–612.
- Gazzani, Andrea, Joseba Martinez, Filippo Natoli, and Paolo Surico. n.d. "The Public Origins of American Innovation."
- Giannetti, Renato, and Michelangelo Vasta. 2006. "Appendix: The Source and the Imita.db Dataset." In *Evolution of Italian Enterprises in the 20th Century*. Heidelberg, GERMANY: Physica-Verlag.
- Giorcelli, Michela. 2019. "The Long-Term Effects of Management and Technology Transfers." *American Economic Review* 109 (1): 121–52.
- Giorcelli, Michela, and Bo Li. 2021. "Technology Transfer and Early Industrial Development: Evidence from the Sino-Soviet Alliance." w29455. National Bureau of Economic Research.
- Huang, Zhangkai, Lixing Li, Guangrong Ma, and Lixin Colin Xu. 2017. "Hayek, Local Information, and Commanding Heights: Decentralizing State-Owned Enterprises in China." *American Economic Review* 107 (8): 2455–78.
- Juhász, Réka. 2018. "Temporary Protection and Technology Adoption: Evidence from the Napoleonic Blockade." *American Economic Review* 108 (11): 3339–76.
- Juhász, Réka, and Nathan Lane. 2024. "The Political Economy of Industrial Policy." *Journal of Economic Perspectives* 38 (4): 27–54.
- Juhász, Réka, Nathan Lane, and Dani Rodrik. 2024. "The New Economics of Industrial Policy." *Annual Review of Economics* 16 (Volume 16, 2024): 213–42.
- Juhász, Réka, Mara P. Squicciarini, and Nico Voigtländer. 2024. "Technology Adoption and Productivity Growth: Evidence from Industrialization in France." *Journal of Political Economy* 132 (10): 3215–59.
- Khwaja, Asim Ijaz, and Atif Mian. 2005. "Do Lenders Favor Politically Connected Firms? Rent Provision in an Emerging Financial Market*." *The Quarterly Journal of Economics* 120 (4): 1371–411.
- Krueger, Anne O., and Baran Tuncer. 1982. "An Empirical Test of the Infant Industry Argument." *The*

- American Economic Review* 72 (5): 1142–52. <https://www.jstor.org/stable/1812029>.
- Lane, Nathan. 2019. “The New Empirics of Industrial Policy.” *Journal of Industry, Competition and Trade*.
- . 2025. “Manufacturing Revolutions: Industrial Policy and Industrialization in South Korea.” *The Quarterly Journal of Economics*, May, qjaf025.
- Monfardini, Patrizio, Paolo Quattrone, and Pasquale Ruggiero. 2022. “Exploring the Middle Way: The Istituto Per La Ricostruzione Industriale (IRI) in Between Corporate Capitalism and Planned Economy (1948–1973).” In *Research in the Sociology of Organizations*, edited by Renate E. Meyer, Stephan Leixnering, and Jeroen Veldman, 77–96. Emerald Publishing Limited.
- Nurkse, Ragnar. 1953. *Problems of Capital Formation in Underdeveloped Countries*. Oxford: Blackwell.
- Rosenstein-Rodan, P. N. 1943. “Problems of Industrialisation of Eastern and South-Eastern Europe.” *The Economic Journal* 53 (210-211): 202–11.
- Semrau, Finn Ole, Sonja Peterson, and Leonie Meissner. 2060. “It’s Not a Sprint, It’s a Marathon: Reviewing Governmental R&D Support for Environmental Innovation” published online; 1-27 (January).
- Toninelli, Pier Angelo, and Michelangelo Vasta. 2011. “Size, Boundaries, and Distribution of Italian State-Owned Enterprise (1939–1983).” In *Reappraising State-Owned Enterprise*. Routledge.
- Toniolo, Gianni, ed. 2013. *The Oxford Handbook of the Italian Economy Since Unification*. 1st ed. Oxford University Press.
- UK Government. 2025. “Industrial Strategy: A 10-Year Plan for Strong, Secure and Sustainable Growth.” Policy Paper. Department for Business; Trade; <https://www.gov.uk/government/publications/industrial-strategy>.
- Vannini, Alessandra. 2017. “Fascist Politics and Autarkic Economy in a Compared Perspective: The Istituto Per La Ricostruzione Industriale (I.R.I.) and the Instituto Nacional de Industria (I.N.I.), 1933-1959.” Thesis, European University Institute.
- Veugelers, Reinhilde, Simone Tagliapietra, and Cecilia Trasi. 2024. “Green Industrial Policy in Europe: Past, Present, and Prospects.” *Journal of Industry, Competition and Trade* 24 (1): 4.
- Wilkinson, Benedict. 2020. “The EU’s Defence Technological and Industrial Base.” Policy Department, Directorate-General for External Policies.